Arunabha Majumder

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Adaptyv Bio, Lausanne

Application: Mechanical / Mechatronics Engineer (Early Career), Hardware

Dear Adaptyv Hiring Team,

I am applying again for the Mechanical / Mechatronics Engineer role, as your email kindly invited. I think my first application did not show clearly enough how hands-on my work is, so I want to fix that here. I enjoy building things and turning ideas into working systems, and helping AI agents run real lab experiments is a problem I would love to work on.

In my PhD at Aalborg University I designed, prototyped and built robotic hardware from concept to testing. For a shoulder exoskeleton, I made custom mechanical components in SolidWorks, 3D printed them, did the mechanical assembly, and integrated a motor, encoder and force sensors. When one clutch part kept breaking, I found the root cause and redesigned it into a simpler part that was easy to fabricate and reliable. The exoskeleton cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I also implemented an Arduino-based trigger system to synchronise a robot with a Qualisys motion capture system, so I know what it takes to make separate instruments work together in one workflow.

Before that, I worked with pneumatics, valves, pressure sensors and data acquisition on pneumatic artificial muscles, and implemented a controller in LabVIEW. In my Bachelor's I did machining and mechanical assembly, and built a Go-Kart and a hybrid vehicle with student teams, so I am comfortable in a workshop with hand and power tools.

My research work is complete, so I can start soon, and I am ready to relocate to Lausanne. I have not built lab automation yet, but I learn quickly and I like working closely with software and biology people to make hardware run reliably with minimal human help. Thank you for taking a second look.

Kind regards,
Arunabha Majumder